

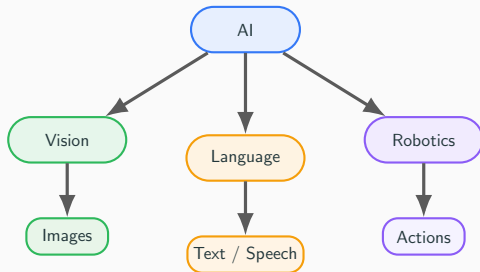
Lecture 1: What Is Artificial Intelligence?

AI Engineering Orientation

Samuel Hailemariam Seifu

Why AI Matters

- AI is now part of daily life
- It powers search, recommendations, speech, and automation
- Modern systems can understand text, images, and audio
- AI is shaping medicine, finance, transport, and education



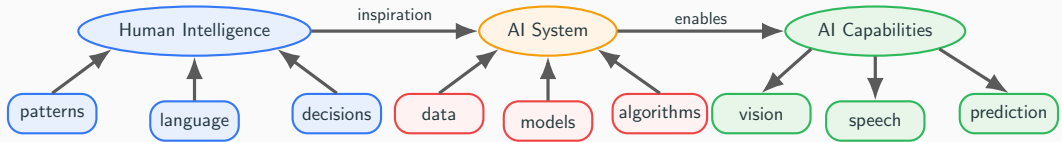
What Is Artificial Intelligence?

Simple Definition

Artificial Intelligence is the field of building systems that can perform tasks that normally require human intelligence.

Human-like tasks

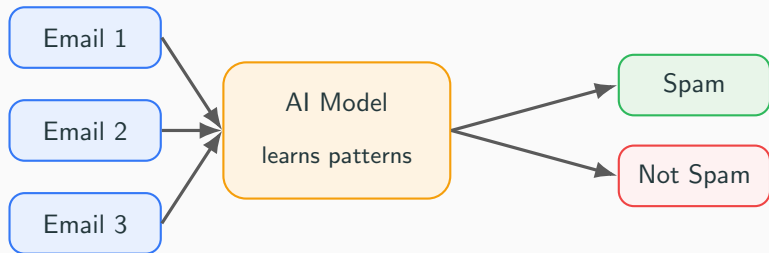
- recognizing patterns
- understanding language
- learning from examples
- making decisions



A Simple Example: Spam Detection

Goal: decide whether an email is spam.

- Humans notice suspicious patterns
- AI learns those patterns from many examples
- After training, it predicts new emails



Narrow AI vs AGI

Narrow AI

- built for one task
- good at a specific problem
- common today

translation

face recognition

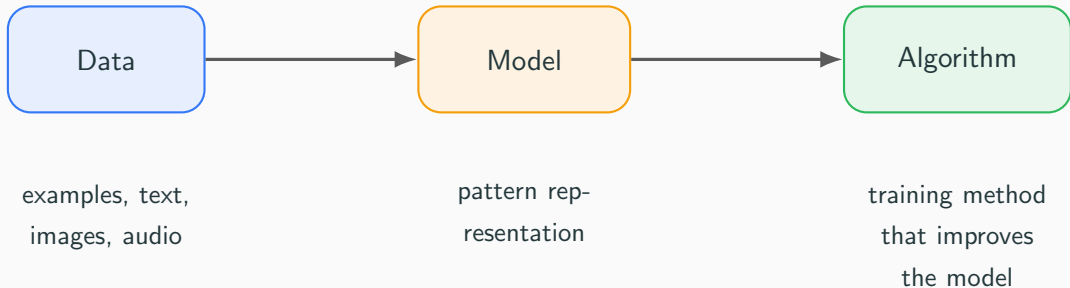
AGI

- would handle many tasks
- would transfer knowledge
- does not exist yet

learn anything

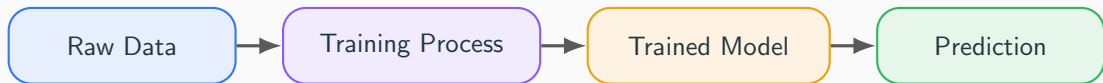
human-level flexibility

How AI Systems Work

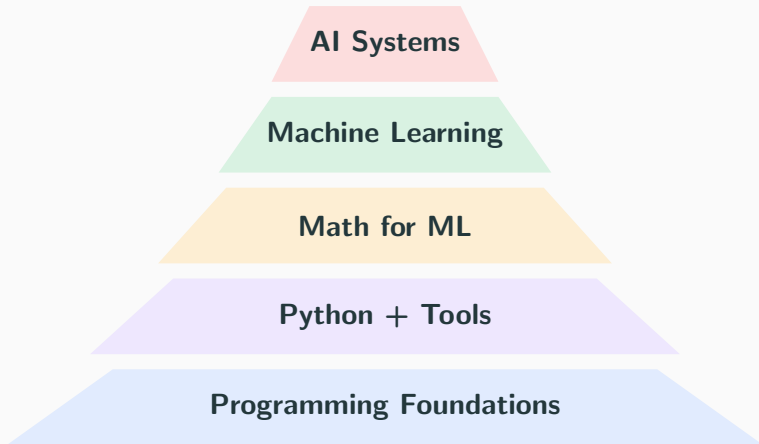


- **Data** gives examples
- **Model** captures patterns
- **Algorithm** adjusts the model to reduce errors

From Data to Prediction



The AI Engineering Stack



Each higher layer depends on the layers below it.

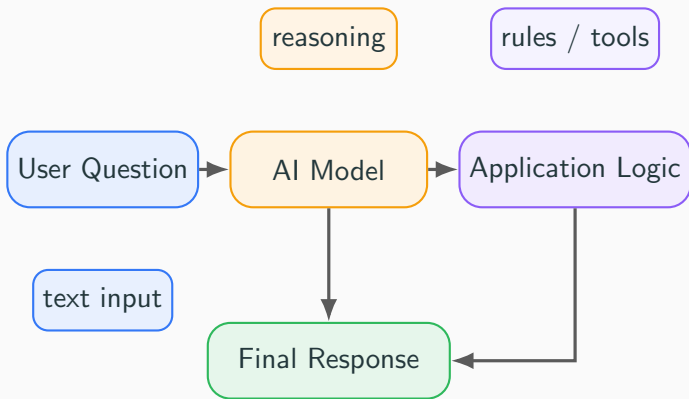
Real AI System Example

Example: an AI chatbot

A real AI product is not just a model.

It includes:

- user input
- model processing
- system logic
- final response



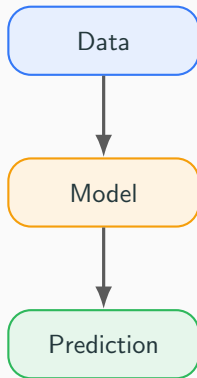
Simple Python Example

```
emails = [  
    "Win a free prize now",  
    "Team meeting at 10 AM"  
]  
  
for email in emails:  
    if "free" in email.lower() or "prize" in email.lower():  
        print("Spam")  
    else:  
        print("Not Spam")
```

- This is a hand-written rule
- Real AI systems learn patterns automatically from data
- Machine learning replaces many fixed rules with learned behavior

Key Ideas From This Lecture

- AI builds systems that perform intelligent tasks
- Most real systems today are **Narrow AI**
- AI depends on **data, models, and algorithms**
- AI engineering combines math, programming, and systems thinking



Next: Types of Artificial Intelligence

Symbolic AI, Machine Learning, and Deep Learning